# Bio SBQ2: Recall Practice

Start time: _____

Topics 1.4 Translation · 1.5 Gene Regulation · 1.6 Cell Specialization

Score: _____ / 15

**How to use:** answer every question **without notes**. Tick the box when you get it right. Check against your practice sheet, then redo only the unticked ones.

## 1.4 Translation

1. Using the genetic code chart, write the amino acid sequence coded by this mRNA, **beginning with the first start codon**. ☐

   `5'-GACCGCAUGGUGACGAAAUUUGGCCAUUAA-3'`

2. The protein-coding part of the *HT1* mRNA is **1170 bases** long. Calculate the length of the wild-type HT1 protein in amino acids. Show your working. ☐

3. In the mutant *ht1¹* allele, nucleotides 795–797 are **AGA** (arginine), replacing the wild-type **lysine**. Predict the 796th nucleotide in the wild-type allele and explain why. ☐

4. A tRNA has the anticodon **3'-UAC-5'**. What is the mRNA codon, and which amino acid does this tRNA carry? ☐

   **mRNA codon:** _____

   **Amino acid:** _____

5. Streptomycin blocks tRNA binding to mRNA in **prokaryotic** ribosomes. Explain its effect on bacteria, **and** why human cells are not affected. ☐

## 1.5 Regulation of Gene Expression

**6** Describe the effect of the **absence** and the **presence** of lactose on RNA polymerase in the *lac* operon. ☐

**No lactose:**

**Lactose present:**

**7** Explain why the *lac* operon is called **inducible**. Then explain **positive control** of the *lac* operon using CAP and cAMP. ☐

**Inducible because:**

**Positive control:**

**8** **What if?** For each case, predict *lac* gene expression (**ON / OFF / LOW**) and give a reason. ☐

| Situation | ON / OFF / LOW | Reason |
|---|---|---|
| a) No repressor is made | | |
| b) Operator mutated so the repressor can't bind | | |
| c) Glucose high **and** lactose present | | |
| d) CAP cannot bind cAMP (glucose low, lactose present) | | |

**9** Describe the role of **phosphorylated p53** after DNA damage. Then predict what happens to the cell cycle if p53 is mutated and **cannot bind DNA**. ☐

**Role of p53:**

**If p53 can't bind DNA:**

**10** Compare DNA methylation and histone acetylation. ☐

| | DNA methylation | Histone acetylation |
|---|---|---|
| **Effect on chromatin** | | |
| **Genes ON or OFF?** | | |

## 1.6 Gene Expression & Cell Specialization

*Use this data for Q11–12. PIG1 codes for the enzyme needed to make purple pigment.*

| Condition | PIG1 mRNA | PIG1 enzyme | Flower |
|---|---|---|---|
| Normal plant | 100 | 100 | Dark purple |
| Activator-binding site mutated | 24 | 22 | Pale purple |
| Repressor-binding site mutated | 165 | 158 | Very dark purple |
| miR-P overexpressed | 96 | 25 | Pale purple |

**11** Explain why overexpressing **miR-P** greatly reduces PIG1 **enzyme** without a similar drop in PIG1 **mRNA**. ☐

**12** Use the data to determine the normal function of the **negative regulatory protein**. Quote numbers. ☐

*Use this data for Q13–14. TF-M activates the TF-C gene. TF-M and TF-C bind together to muscle genes.*

| Treatment | Structural gene | Metabolic gene | Signaling gene |
|---|---|---|---|
| None added | 3 | 5 | 4 |
| TF-M added | 38 | 34 | 31 |
| TF-C added | 8 | 10 | 9 |
| TF-M + TF-C | 94 | 91 | 88 |
| TF-M + TF-C + negative regulator | 19 | 22 | 18 |

**13** Describe the evidence that TF-M and TF-C regulate muscle genes through **combinational control**. ☐

**14** Predict the effect of a mutation that stops TF-M from binding to the regulatory region of the **TF-C gene**. Justify using the data. ☐

15 Muscle cells and stem cells contain the **same DNA**. Explain how they end up with different structures and functions. ☐

## Calc 7.8–7.9: Templates (5 min)

16 Write the general solution of **dy/dt = ky**. ☐

**y =** _____

17 For **dP/dt = kP(1 − P/L)**: ☐

**a) Limit of P as t → ∞:** _____

**b) P is growing fastest when P =** _____

Bio right: _____ / 15 Calc right: _____ / 2 Questions to redo: _____